Compile on the PC. Execute on the controller.

IMPLEMENTED ON THE NUCLEO BASELINE

Native execution, serial download and explicit activation; coherent monitoring; compatible state migration, trial and rollback; TON on-delay timers.

A1 INTEGRATION STILL TO DO

Standalone clock and pin setup, four-channel I/O validation, OLED driver and heartbeat ownership. Native USB and Ethernet are not ready-to-use A1 features.

The supervisor owns physical I/O and fault handling. User logic runs in a restricted native worker. The UI must consume snapshots at lower priority, with bounded I²C transactions.

Target scan period: 10 ms. Existing measured evidence is for NUCLEO-F446RE; A1 timing must be measured separately.